CHAPTER 12

Parametric representations and Cartesian equations

S65, S67, S70

Before you begin · Lines, planes and systems

Without a calculator, in six minutes. Answer briefly. Use direction vectors, normal vectors and membership. Then choose the targeted follow-up indicated under the questions.

1. Give a direction vector of (AB) for A=(1;2;3), B=(3;1;3).
2. Check that (1;2;-1) is orthogonal to (2;-1;0) and (1;0;1).
3. Is the vector (1;1;1) orthogonal to (1;2;-1)?
4. Solve 2 $t = 6$.
5. What must be checked for a point to be the projection of M onto a plane?
6. Are the vectors (1;0;0) and (0;1;0) collinear?

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Subtract A from B, coord. by coord.; R05. **2.** Calculate the two scalar products separately; R60. **3.** Add the three products and compare to zero; R60. **4.** Divide both sides by 2 and check. **5.** Go back over the two distinct conditions for the projection; R63. **6.** Compare the first then the second coordinates; R06. The R questions serve as review prompts with the teacher, even if their collective routine is planned for later.

Objectives

Describe a line and a plane; find intersections; choose a method for a projection.

Course notes

Throughout this chapter, coordinates are given in an orthonormal coordinate system of space. This assumption is necessary to identify a normal vector from the coefficients of an equation and to calculate distances.

Parametrised lines

The line through $A(x_A, y_A, z_A)$ with non-zero direction vector (a, b, c) has the representation

$$d : \begin{cases} x = x_A + at \\ y = y_A + bt \\ z = z_A + ct \end{cases} \quad (t \in \mathbb{R}).$$

A point lies on the line if there exists one real t satisfying all three equalities. A line has many representations: changing the starting point or multiplying the direction vector by a non-zero real number.

Example. $(x,y,z) = (1,2,0) + t(2,-1,3)$ describes the same line as $(3,1,3) + s(4,-2,6)$; the link is $t = 1 + 2s$.

Plane equations

Equation of a plane

If $(a,b,c) \neq (0,0,0)$, the set $ax + by + cz + d = 0$ is a plane with normal vector (a,b,c). The plane with normal $\vec{n}$ through A is constructed by writing $\overrightarrow{AM} \cdot \vec{n} = 0$.

Proof. The condition $\overrightarrow{AM} \cdot \vec{n} = 0$ is written $a(x - x_A) + b(y - y_A) + c(z - z_A) = 0$. Expanding, $d = -ax_A - by_A - cz_A$. Conversely, every point satisfying this equation satisfies the orthogonality condition, so it belongs to the plane.

Example. The plane with normal $(2,-1,3)$ through $A(1,2,0)$ is $2(x-1) - (y-2) + 3z = 0$, i.e. $2x - y + 3z = 0$.

Intersections

For a line and a plane, replace x, y, z in the equation of the plane by their parametric expressions. A solution in t gives a single point of intersection; an identity gives a line contained in the plane; a contradiction gives a disjoint parallel line. For two planes, non-collinear normal vectors give an intersection along a line; solve the system of two equations by choosing one variable as a parameter. If the normal vectors are collinear, the planes are parallel: distinguish coincident from disjoint. For two lines, use parameters s, t (independent) and solve the three equations. With no solution and non-collinear direction vectors, the lines are non-coplanar.

Projection onto a plane

If $\mathcal{P} : ax + by + cz + d = 0$, find the projection of M on the normal line through M : $H = M + t(a,b,c)$. Substitute into the equation of the plane to determine t, then calculate MH. We deduce, without making it a formula to be learnt in isolation,

$$d(M,\mathcal{P}) = \frac{|ax_M + by_M + cz_M + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Checking is essential. After calculating a projection, check simultaneously $H \in \mathcal{P}$ and the collinearity of $\overrightarrow{MH}$ with the normal vector.

Moving from objects to equations

12.1 One line, several parametrisations

First use: S65.

Definition · Parametrising a line

A line through A with non-zero direction vector $\vec{u}$ is the set of points M such that $\overrightarrow{AM} = t\vec{u}$, with $t \in \mathbb{R}$. The parameter is neither a prescribed coordinate nor a time, unless the context specifies it.

Worked example · Same set, other parameters

The expressions $(1+t;2-t;2t)$ and $(3+2s;-2s;4+4s)$ describe the same line: set $t = 2+2s$ transforms the first into the second, and this transformation runs through all real numbers. The starting points may differ, and the direction vectors may be proportional.

A1 · Immediate application Show that $(t;1+2t;3-t)$ and $(2+3s;5+6s;1-3s)$ describe the same line. Is the point $(1;3;2)$ belong to it?

12.2 A normal vector fixes the equation of a plane

First use: S67.

Property · Cartesian equation

In an orthonormal coordinate system, a plane with nonzero normal vector $(a;b;c)$ through $A = (x_A;y_A;z_A)$ has equation $a(x-x_A)+b(y-y_A)+c(z-z_A)=0$. It expresses $\overrightarrow{AM}\cdot\vec{n}=0$.

Worked example · Construct, then test

For A=(1;0;2) and $\vec{n} = (2;-1;3)$, the plane has equation 2 $x-y+3z-8=0$. The point (0;1;3) lies in the plane since 0 $-1+9-8=0$. A point does not lie in the plane just because its triple resembles the normal vector.

A2 · Partial guidance Determine the plane with normal vector $(1;2;-1)$ through (2;0;1). Test whether (1;1;2) lies in it, then give two non-collinear directions of the plane.

12.3 Three outcomes for a single equation

First use: S69.

After substitution	Interpretation	Example
$at+b=0,\ a\neq 0$	a point	intersecting line
$0t+b=0,\ b\neq 0$	no point	strictly parallel
$0t+0=0$	all points	line in the plane

Worked example · Reading the equation y = mx + p

For the plane x+y+z=1 and the line $(t;2t;1-t)$, substitution gives 2 $t+1=1$, so t=0 and a single intersection (0;0;1). We conclude only after substituting the value back into all three coordinates.

A3 · Compare In the plane $z=1$, study the relative positions of the lines $d_1 : (t;0;t)$, $d_2 : (t;0;0)$ and $d_3 : (t;0;1)$. Give the intersection point when it is unique and sketch the three situations.

12.4 Two lines, two independent parameters

First use: S70.

Worked example · Do not impose a false simultaneity

For $d : (t;0;0)$ and $d' : (1;1+s;0)$, the intersection requires t=1 and s=-1. Imposing t=s would wrongly eliminate the point (1;0;0). The parameters are used separately to run along each line.

A4 · Fix a method For $d : (t;t;0)$ and $d' : (1+s;1-s;s)$, solve with two parameters, then explain why all three coordinates must be checked.

12.5 Two checks after the system

First use: S70.

Worked example · Projecting onto a plane

Let $\mathcal{P} : x+2y-z=1$ and M=(1;1;1). The normal line through M is $H=(1+t;1+2t;1-t)$. Substituting into the plane gives 2 $+6t=1$, therefore $t=-1/6$. H=(5/6;2/3;7/6). We check 5 $/6+4/3-7/6=1$ and $\overrightarrow{MH}=(-1/6)(1;2;-1)$. The distance is $\overline{6}/6$.

A5 · Synthesis with a degenerate case For a real number m, d_m passes through (0;0;m) and has direction $(1;m;1)$. Study the intersection with $x+2y-z=1$. Treat m=0 separately. Is there an m for which the line is perpendicular to the plane?

A first step: Substitute the three coordinates before dividing. For orthogonality, compare the direction vector with a normal vector.

A6 · Choice of strategy Determine the intersection of the planes $x+y+z=3$ and $x-y=0$. Give a point and a direction vector, then check these two pieces of information in the equations.

One minute, spoken aloud. State the two properties that make it possible to recognise a projection without mentioning coordinates.

Delayed review · S71 then S87. Match each line of an algebraic calculation with a geometric property; deal first with the parameters that make a denominator zero.

Investigation

Three lines and a plane - 30 min.

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x+y+z=3$. In each question, the parameter t runs through $\mathbb{R}$.

1. The line d_1 has the representation $(x;y;z)=(t;t;t)$. Determine $d_1 \cap \mathcal{P}$.

2. The line d_2 has the representation $(x;y;z) = (3+t;-t;0)$. Determine $d_2 \cap \mathcal{P}$.

3. The line d_3 has the representation $(x;y;z) = (t;-t;0)$. Determine $d_3 \cap \mathcal{P}$.

4. For each line, state whether the equation obtained after substituting into the plane's equation has a unique solution, no solution or every real number as a solution. Relate these cases to the geometric positions found.

Progressive exercises

E1 · Direct application. In a coordinate system of space, consider $A(1;-1;2)$ and $B(3;0;1)$. Give a parametric representation of the line (AB).

E2 · Application and justification. In an orthonormal coordinate system, determine a Cartesian equation of the plane passing through $A(1;2;3)$ with normal vector $\vec{n} = (1;0;-2)$.

E3 · Method to organise. In a coordinate system of space, the line d is given by $(x;y;z) = (t;1+t;2-t)$, $t \in \mathbb{R}$, and the plane $\mathcal{P}$ by $x+2y+z=8$. Determine their intersection.

E4 · Choice of strategy. In a coordinate system of space, determine the intersection of the planes with equations $x+y+z=1$ and $x-y=0$. Give a parametric representation if the intersection is a line.

E5 · Synthesis or proof. In a coordinate system of space, study the relative position of the planes with equations 2 $x-y+z=3$ and $4x-2y+2z=5$. Justify.

Work from the textbook or a past paper

Baccalauréat, Metropolitan France, 17 June 2025: exercise 3, statements 1 to 4 (p. 5-6). At home after S70. Justify each answer; for a false statement, explain precisely what disproves it. The paper is available here: official past paper . Use the data and instructions of this past paper.

Homework

DM12 - Projection and intersection - 55 min.

In an orthonormal coordinate system of space, consider the point $M(3;0;0)$ and the planes $\mathcal{P} : x+2y+2z = 9$ and $\mathcal{Q} : x-y=0$.

1. Give the coordinates of a nonzero normal vector to $\mathcal{P}$.

2. Determine the coordinates of the orthogonal projection H of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$.

3. Show that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it.

4. The point H lie on this line? Justify.

5. Determine the coordinates of the point K of this line whose z-coordinate is zero. Calculate MK, then compare MK and MH.

MCQ A · reading and checking a representation

Orthonormal coordinate system. After S67; MCQ B after S70.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. $d : (x;y;z) = (1;0;2) + t(2;-1;0)$ has direction vector:

A. $(1;0;2)$ B. $(2;-1;0)$

C. $(2;1;0)$ D. $(0;0;2)$

2. Which of these points belongs to d ?

A. $(3;1;2)$ B. $(1;0;0)$

C. $(2;-1;0)$ D. $(3;-1;2)$

3. A normal vector to the plane $x + 2y - 2z = 3$ is:

A. $(1;2;-2)$ B. $(1;2;3)$

C. $(3;0;0)$ D. $(-2;1;0)$

4. To check that H is proj. of M on $\mathcal{P}$, we need:

A. only $H \in \mathcal{P}$ B. only MH small

C. $H \in \mathcal{P}$ and $\overrightarrow{MH}$ normal to the plane D. that $H = M$

5. The line $A + tu$ with $u \neq 0$ is also represented by:

A. $A + t\vec{0}$ B. $A + 2u + s(3u)$

C. $tA + u$ D. the single point $A + u$

MCQ B · intersections and special cases

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. Substituting a line into a plane equation gives $0 = 4$. The intersection is:

A. a line B. a point

C. the plane D. empty

7. If substitution gives $0 = 0$ for every parameter value:

A. the line lies in the plane B. there is no common point

C. there is exactly one common point D. the data are wrong

8. To find the intersection of $A + tu$ and $B + sv$, we must:

A. set $s = t$ B. replace both parameters with 0

C. solve for s, t independent D. ignore the third coordinate

9. The intersection of $x + y = 1$ and $z = 2$ is:

A. a plane B. the line $(t;1-t;2)$

C. the point $(1;0;2)$ D. empty

10. For $m = 0$, the set $mx + y = 1$ is:

A. empty B. a line

C. the whole of space D. the plane $y = 1$

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Two lines whose direction vectors are not collinear necessarily intersect.

2. The planes $x + y + z = 1$ and $2x + 2y + 2z = 3$ coincide.

3. The projection of M onto a plane is unique.